WBKMNet: A Novel Wavelet-Boundary KAN-Mamba Network for Boundary-Critical Medical Image Segmentation

Objective

①Medical image segmentation is vital for CAD, yet ambiguous-boundary lesions are challenging.

②KAN-augmented Mamba models attain promising accuracy–efficiency balance.

③They suffer three boundary-fidelity limitations: lost high-frequency details, ignored directional anisotropy, and output-only supervision.

Results & Conclusion

On the CVC-ClinicDB, GlaS, and BUSI benchmarks, WBKMNet attains the highestDice and Intersection-over-Union among the compared state-of-the-art methods, surpassing thestrongest competitor by up to 2.63 percentage points.

WBKMNet yields sharper,better-localised boundaries and strengthens the reliability of downstream area and marginmeasurements.

Methods

We propose the Wavelet-Boundary KAN-Mamba Network (WBKMNet), which comprises three modules:

①a wavelet-KAN patch em-bedding that performs near-lossless four-fold downsampling via a two-level Haar decompositionand band-specific KAN encoders;

②a KAN-gated multi-path selective scan with a token-wise KAN gate that adaptively weights the four scan directions;

③a boundary-awaredeep supervision scheme with a Sobel-driven KAN edge gate and multi-scale auxiliary heads.

